

HDOC DAY – 32

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Batch-15

QUESTION:

Write a C program to perform the following menu-driven operations using separate functions:

1. Get two variable length 2D arrays and prove that matrix addition is commutative.
2. Get two variable length 2D arrays and prove that matrix multiplication is not commutative.
3. Exit.

Aim

To write a C program to perform matrix addition and matrix multiplication using separate functions and verify their commutative properties.

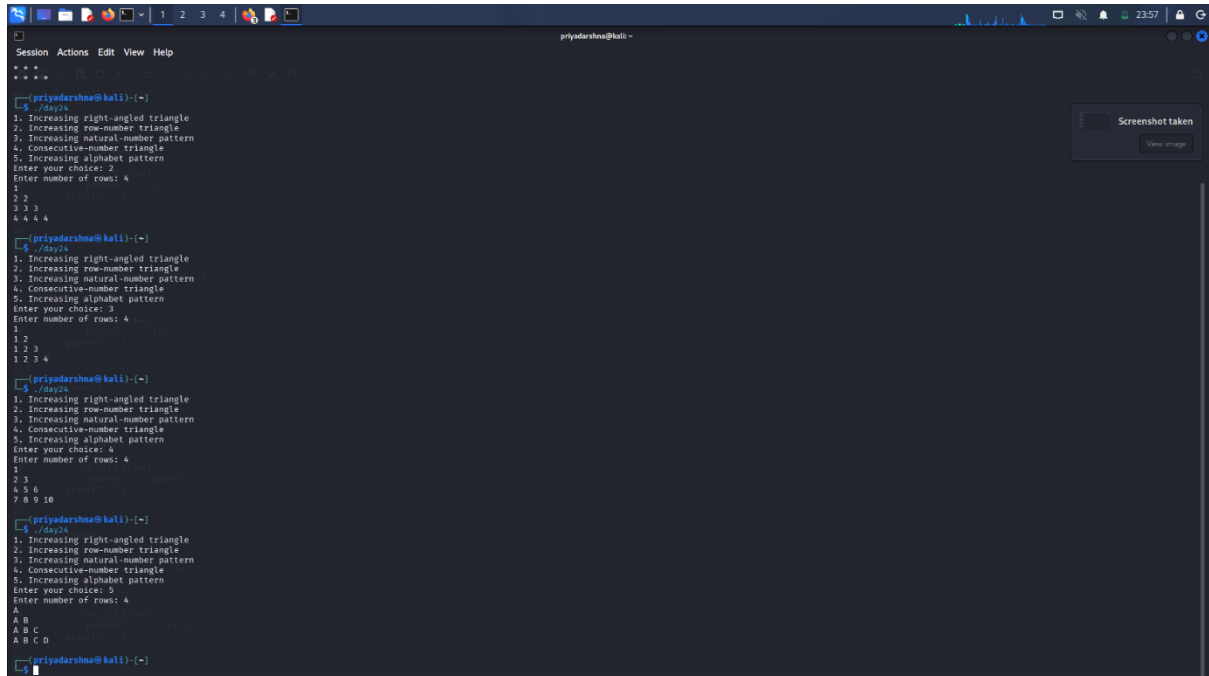
Algorithm

1. Start.
2. Display the menu for addition, multiplication and exit.
3. Read the user's choice.
4. Enter two matrices.
5. If choice is 1, calculate $A+B$ and $B+A$.
6. Compare the results to verify addition is commutative.
7. If choice is 2, calculate $A \times B$ and $B \times A$.
8. Compare the results to verify multiplication is not commutative in general.
9. If choice is 3, exit.
10. Stop.

>C Program:

```
1#include <stdio.h>
2
3void pattern1();
4void pattern2();
5void pattern3();
6void pattern4();
7void pattern5();
8
9int main()
10{
11    int choice;
12
13    printf("1. Increasing right-angled triangle\n");
14    printf("2. Increasing row-number triangle\n");
15    printf("3. Increasing natural-number pattern\n");
16    printf("4. Consecutive-number triangle\n");
17    printf("5. Increasing alphabet pattern\n");
18
19    printf("Enter your choice: ");
20    scanf("%d", &choice);
21
22    switch(choice)
23    {
24        case 1: pattern1(); break;
25        case 2: pattern2(); break;
26        case 3: pattern3(); break;
27        case 4: pattern4(); break;
28        case 5: pattern5(); break;
29        default: printf("Invalid choice");
30    }
31
32    return 0;
33}
34
35void pattern1()
36{
37    int n,i,j;
38    printf("Enter number of rows: ");
39    scanf("%d", &n);
40    for(i=1; i<=n; i++)
41    {
42        for(j=1; j<=i; j++)
43            printf("%d ", j);
44        printf("\n");
45    }
46}
47
48void pattern2()
49{
50    int n,i,j;
51    printf("Enter number of rows: ");
52    scanf("%d", &n);
53    for(i=1; i<=n; i++)
54    {
55        for(j=1; j<=i; j++)
56            printf("%d ", i);
57        printf("\n");
58    }
59}
60
61void pattern3()
62{
63    int n,i,j;
64    printf("Enter number of rows: ");
65    scanf("%d", &n);
66    for(i=1; i<=n; i++)
67    {
68        for(j=1; j<=i; j++)
69            printf("%d ", j*j);
70        printf("\n");
71    }
72}
73
74void pattern4()
75{
76    int n,i,j,num=1;
77    printf("Enter number of rows: ");
78    scanf("%d", &n);
79    for(i=1; i<=n; i++)
80    {
81        for(j=1; j<=i; j++)
82            printf("%d ", num++);
83        printf("\n");
84    }
85}
86
87void pattern5()
88{
89    int n,i,j;
90    printf("Enter number of rows: ");
91    scanf("%d", &n);
92    for(i=1; i<=n; i++)
93    {
94        for(j=1; j<=i; j++)
95            printf("%c ", 'A'+j-1);
96        printf("\n");
97    }
98}
99
```

>OUTPUT



```
(priyadarshna@kali)~$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4

(priyadarshna@kali)~$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4

(priyadarshna@kali)~$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10

(priyadarshna@kali)~$ ./day24
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows: 4
A
A B
A B C
A B C D

(priyadarshna@kali)~$
```

>TEST CASE TABLE:

Test Case	Choice	Matrix Size	Result
1	1	2×2	Addition is commutative
2	2	2×2	Multiplication is not commutative
3	3	-	Exit

>RESULT:

Thus, the C program to perform matrix addition and matrix multiplication using separate functions was executed successfully, and their commutative properties were verified.